

Adrian Nguyen

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Education

- Edison High School, Huntington Beach, CA – Junior | Graduation Date: June 2025. Rank: 5 of 450
 - GPA Weighted Total: 4.67; Weighted Grades 10-12: 4.80; Unweighted: 4.0
 - SAT 1570/1600; 770 English 800 Math | PSAT: 1450/1520
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Courses / Projects / Experience

Relevant Courses

- AP Courses – AP Computer Science A (5) AP Physics (5), AP Calculus B/C (5), AP Chemistry (5)
- Other APs – AP Human Geography (5), AP European History (5), AP Lang & Comp (4), AP Spanish (4), AP Art History

Research

- Research Assistant Intern for Professor Glenn Healey - EE and CS and Director of Computer Vision Lab at UC Irvine
 - Developed an algorithm to represent orientation of a baseball to longitude/latitude across its surface
 - Manipulated the values and direction of 3D vectors using matrices. Utilized Rodrigues' rotation formula to translate and rotate vectors. Wrote scripts in MATLAB for implementation and efficiency
 - Used Access database and SQL to run queries for data analysis. Compiled percentile tables, created refined datasets for targeted analysis
 - Used MATLAB to create scatterplots for analysis – helped interpret data from a new multi-camera ball tracking system; helped UCI Baseball team achieve their best ever record (45-14)
 - Utilized Python to develop an CatBoost regression model to develop gradient heatmaps using R. Implemented GridSearch algorithm to find the best parameters for the model.
- Orange County Science and Engineering Fair – Recipient of two Special Awards
 - Project Name: Deep Learning for the Prediction and Analysis of Water Pollution Trends
 - Competed against about 500 high school students across Orange County, CA
 - Created Neural Network models to predict the potability of water based on 9 independent variables (pH, Hardness, Sulfate concentration, etc.)
 - Made predictions on a comprehensive dataset from Kaggle
 - Received the Special Awards of the Stockholm Junior Water Prize from the California Water Environment Association (Qualified for State Competition) and the Excellence in Water from the Orange County Water District

Personal Projects

- Developed MindTrain - Brain Games – iOS app to increase brain stimulation for seniors with degenerative diseases (11th, 12th Grade)
 - Designed using Swift, the app provides brain-stimulating exercises with visually pleasing user interfaces for seniors with degenerative diseases based on Alzheimer's research, focusing on easy use for seniors
 - Implemented NavigationLink and modular architecture
 - Currently 2,000+ impressions, 5-star average rating
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Work Experience

- IT/Networking Intern at In Home Doctors Inc. (11th and 12th Grade)
 - Managed/configured office network and ensure functionality of hardware
 - Troubleshoot technical issues and performed backups of internal databases

Competitions / Awards

- American Computer Science League Champion – First Place Award
 - Top 2.1% of all High School competitors internationally in Senior 3/Senior 5 Division
 - Highest score in-state and internationally (40/40 Points)
 - Required general programming knowledge and familiarity with core computer science concepts like Boolean Algebra, LISP, Data Structures, and Assembly
 - USA Coding Olympiad – Silver Tier Qualifier
 - Solved programming problems that involve knowledge of time complexity, advanced data structures, sorting, and greedy algorithms
 - AFA CyberPatriot National Competition – Platinum Tier (Top 30% of Teams) – 2021, 2022, 2023
 - Competed in teams of 5, securing Linux, Windows, Windows Server OS
 - Strengthened security protocols and removed malicious software
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Leadership / Extracurricular Activities

- EHS CyberPatriot Club - President
 - Hosted mock competitions securing Windows, Windows Server, and Ubuntu (Linux) Operating Systems
 - Won sponsorship from Coastline Community College
 - Math Club – Vice President
 - Participated in teams in the OCC Math Meet, M3 Modeling Challenge, and the California Mathematics League competitions
 - DYLAN Non-Profit Organization – Co-Founder & Chief Technology Officer (Involved 11th, 12th Grade)
 - Fundraise to organize social events for special-needs individuals to improve their self-esteem and confidence
 - Developed website for awareness/advertisement
 - Raised \$2500
 - Boy Scouts of America – Eagle Scout
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Skills / Knowledge

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| • Java, Python, SQL, MATLAB | • National/State Competitions |
| • Windows/Linux OS & Networking | • Leader & Team Player |
| • Microsoft Office & Access | • Strong Communicator |